Milestone 5 Documentation

Milestone 5 was basically about getting the power system model ready for actual power flow analysis instead of just modeling connections between parts of the system. In this milestone, a Settings class was added, the Bus class was updated, and the Generator and Load classes were changed so they could better represent what is happening in the system. These updates help set up the project for the next milestone, where the real power flow calculations will happen.

The new Settings class is there to hold system-wide values that the rest of the program can use. It includes the system frequency and the base apparent power, with default values of 60 Hz and 100 MVA. This makes things easier because instead of putting those values in different places throughout the code, they are all stored in one class. That helps keep the per-unit setup consistent.

The Bus class was changed to include more of the information needed for power flow. It now stores the voltage magnitude, voltage angle, and the bus type. The valid bus types are Slack, PQ, and PV, and the class is supposed to raise an error if something else is entered. This is important because each type of bus is treated differently in power flow problems, so the model needs to keep track of that correctly.

The Generator and Load classes were also updated so they can represent power in per-unit form. The Generator class now models real power injection, while the Load class models both real and reactive power consumption. Each class includes methods to calculate those values. These changes matter because power flow depends on knowing how much power is being supplied and used at each bus.